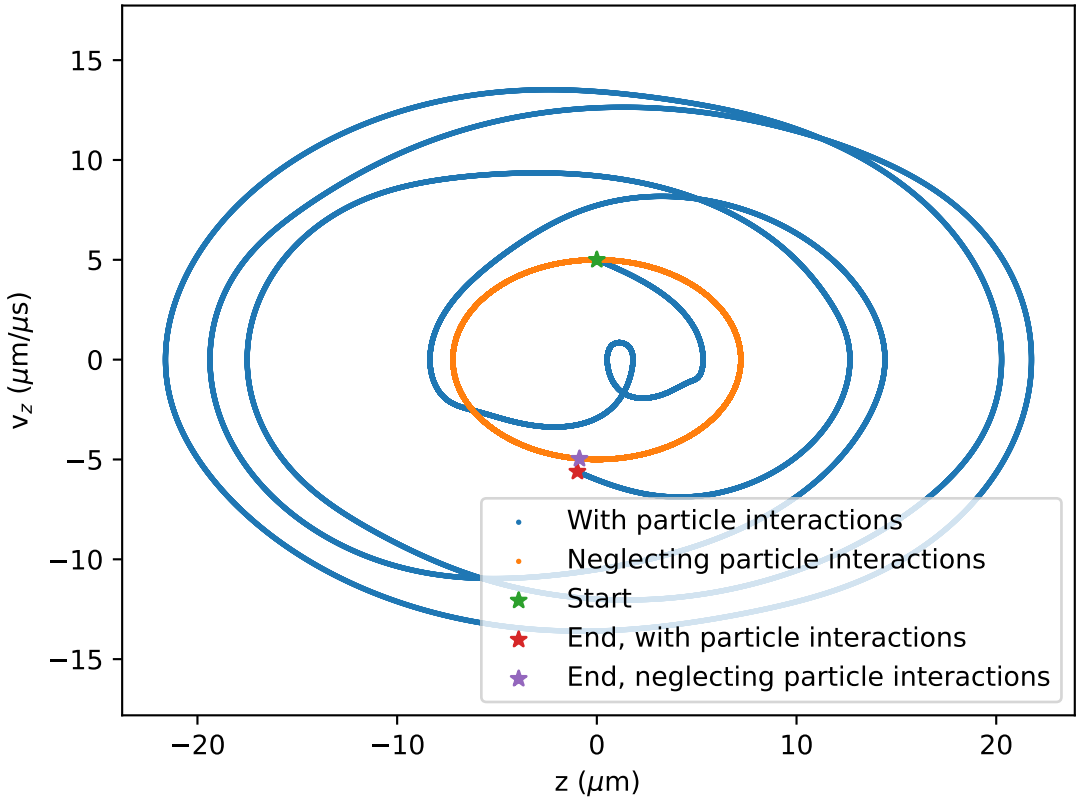

Phase space z, with and without particle interactions, RK4
15
10
5
0
−5
−10
−15
v_z (μm/μs)
−20
−10
0
10
20
z (μm)
With particle interactions
Neglecting particle interactions
Start
End, with particle interactions
End, neglecting particle interactions